- Azure Monitor Logs (http://mng.bz/MBGQ) allows you to collect, organize, and analyze log and performance data from monitored resources in Azure (from resource groups to the app level). It organizes logs in workspaces, with queries written in Kusto Query Language (KQL), which is somewhat similar to what you find in CloudWatch Logs Insights. It sports an interesting feature called *functions*, enabling you to reuse query logic as well as rich export functionality (rule-based with various destinations, from storage to event hub).
- Google Cloud Logging https://cloud.google.com/logging/) is part of the Google Cloud's operations suite, and it includes storage, query, and log routing. As the top-level unit, Google Cloud Logging uses the log to organize entries. The log comprises log entries (events) that can stem from Google Cloud services or third parties (platform-level) as well as applications. What stands out in this context is how flexible the destinations for log routing are: from JSON files stored in Cloud Storage buckets (think the equivalent of S3 buckets) to BigQuery (a large-scale, ad hoc SQL distributed query engine for data warehousing) to Pub/Sub (Google's event stream) to things that are more comparable with what other clouds provide—that is, log "buckets" that support customizable retention periods.

Let's have a look at a concrete logs example using CloudWatch. Examine the following CloudWatch Logs Insights query (where CloudWatch reports it scanned 3.7 GB of data, covering 3.1 million records):

```
fields @timestamp, @message
| sort @timestamp desc
| limit 25
```

Selects fields @timestamp and @message for output; we expect to see a table with two columns as a result.

Sorts by time, descending, or more simply put, most recent event first

Only shows 25 results in the output

WARNING With the `limit 25` in the preceding CloudWatch Logs Insights query example, we instructed CloudWatch to limit the output displayed to 25, but CloudWatch still needs to scan a large number of logs, depending on the time range selected. The time range, and with it the volume of the data scanned, is something you should always consider. On the one hand, it can take a long time, up to several minutes, for the query to return, which is far from interactive usage; and on the other hand, you're oftentimes charged for the amount of data that has been scanned.

Figure 5.1 shows the result of the preceding CloudWatch Logs Insights query. Typically, cloud providers land all the system-level logs (sometimes called *platform logs*, such as syslog and journald but also cloud-platform/API logs) in their own offerings, which makes it a good default starting point, especially for all nonapplication-level logs. If you're curious how to route your application-level logs to these destinations, refer to